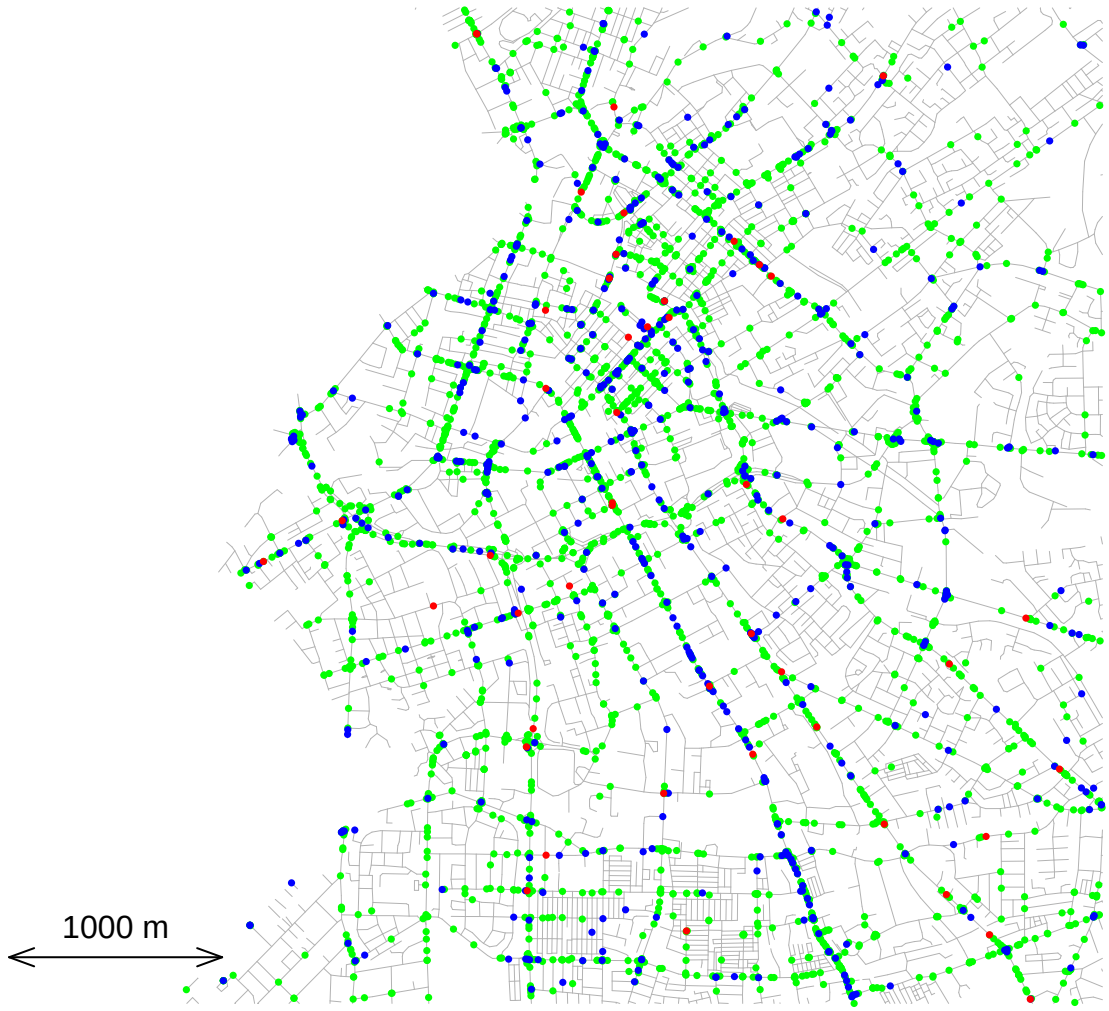
Manchester_0313

```
Wsub <- owin(xrange = c(382000, xbreak[3]),
                 yrange = c(395000, ybreak[4]))

X_sub <- vector("list", 3) # X_sub
for (i in seq_len(3)){
  X_sub[[i]] <- XSeverity_manCity[[i]][Wsub, clip = TRUE, drop = TRUE]
}


dir.create("~/Library/Mobile Documents/com~apple~CloudDocs/Rstudio/0313Fri/CM_data", showWarnings = FALS
# save
saveRDS(XSeverity_manCity, file = "~/Library/Mobile Documents/com~apple~CloudDocs/Rstudio/0313Fri/CM_da


#
L <- domain(X_sub[[1]])

op <- par(mar=c(1,1,1,1))
plot(L, main="Accident with different severity on Manchester city center", col="grey70", lwd=0.4)


plot(as.ppp(X_sub[[3]]), add=TRUE, cols="green", pch=16, cex=0.5)
plot(as.ppp(X_sub[[2]]), add=TRUE, cols="blue", pch=16, cex=0.5)
plot(as.ppp(X_sub[[1]]), add=TRUE, cols="red", pch=16, cex=0.5)



usr <- par("usr")  # c(xmin, xmax, ymin, ymax)

x0 <- usr[1] + 0.08*(usr[2]-usr[1])
y0 <- usr[3] + 0.08*(usr[4]-usr[3])

sb <- spatstat.geom::yardstick(
  x0 = x0, y0 = y0,
  x1 = x0 + 1000, y1 = y0,   # 1000m measuring scale
  txt = "1000 m"
)

plot(sb, add = TRUE)
```




```r
par(op)
```

```r
X1 <- unmark(X_sub[[1]])
X2 <- unmark(X_sub[[2]])
X3 <- unmark(X_sub[[3]])


Y <- superimpose(sev1 = X1, sev2 = X2, sev3 = X3)
# make sure the marks are factors
marks(Y) <- factor(marks(Y), levels = c("sev1","sev2","sev3"))

table(marks(Y))
```

```
##
## sev1 sev2 sev3
##   47  572 3350
```

```r
r_grid <- seq(0, 5000, length=101)

K12 <- linearKcross(Y, i="sev1", j="sev2", r=r_grid)
K21 <- linearKcross(Y, i="sev2", j="sev1", r=r_grid)
K13 <- linearKcross(Y, i="sev1", j="sev3", r=r_grid)
K31 <- linearKcross(Y, i="sev3", j="sev1", r=r_grid)
K23 <- linearKcross(Y, i="sev2", j="sev3", r=r_grid)
K32 <- linearKcross(Y, i="sev3", j="sev2", r=r_grid)
```

```r
mean(nndist(X1))
```

```
## [1] 335.7563
```

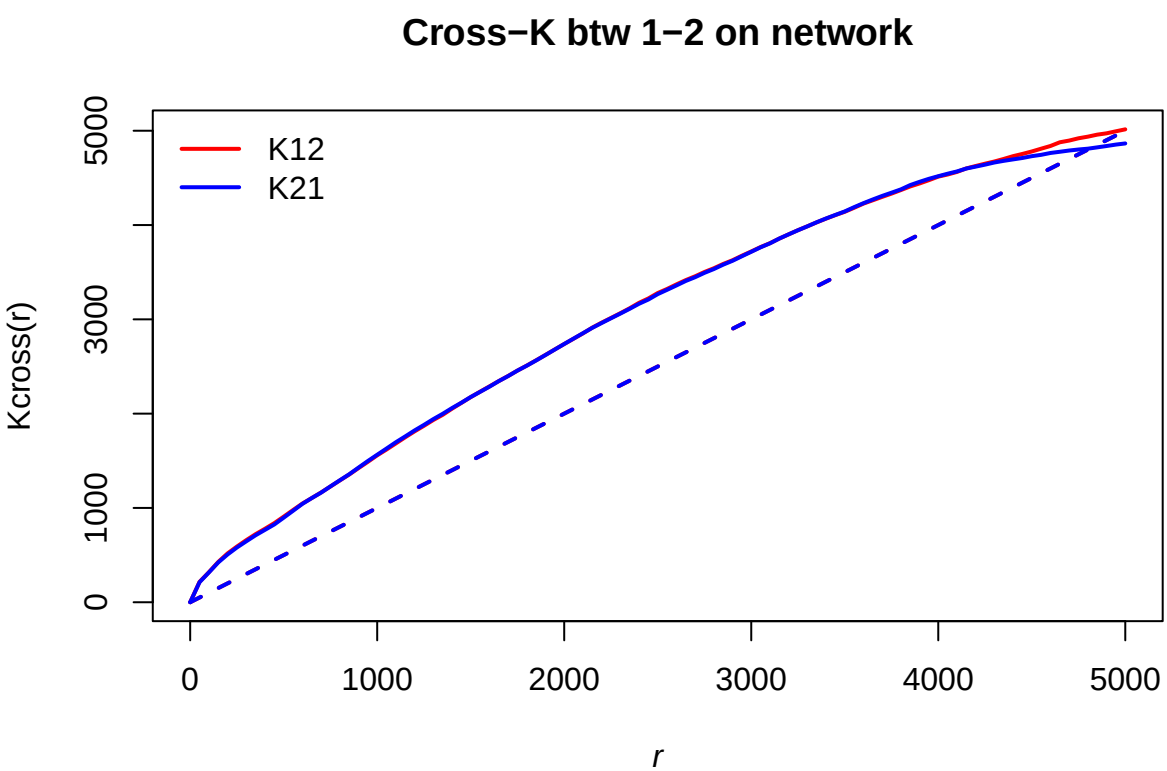
```r
Window(X1)
```

```
## window: rectangle = [382000, 386510.7] x [395000, 399673.3] units
```

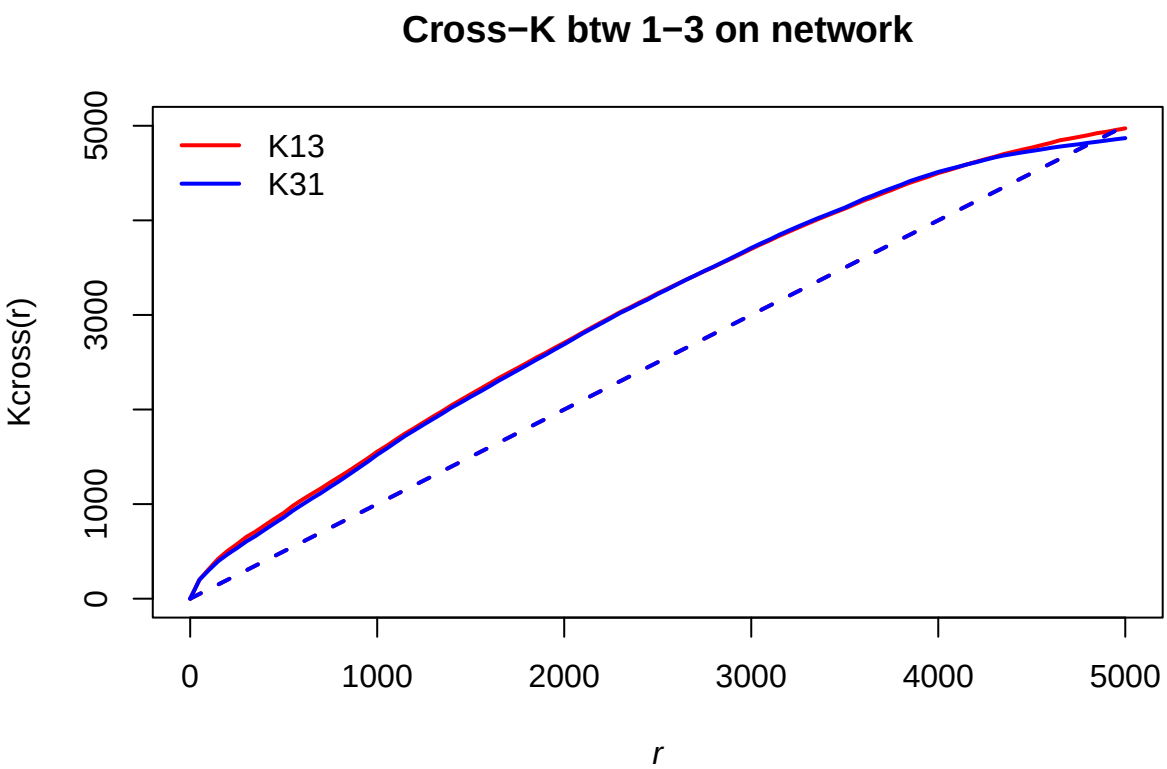
```r
W1 <- boundingbox(X1)
W1
```

```
## window: rectangle = [382572.2, 386308.8] x [395020, 399552] units
```

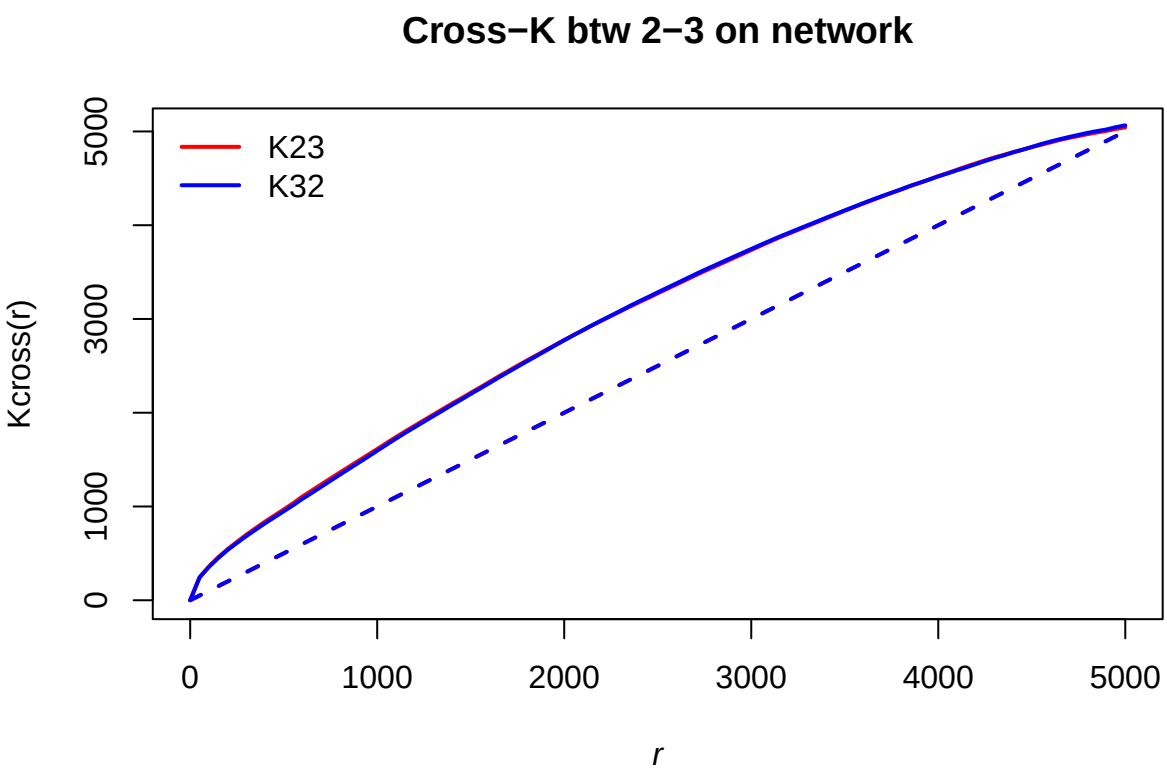
```r
plot(K12, legend=FALSE, ylab = "Kcross(r)", lwd=2, col="red", main="Cross-K btw 1-2 on network")
plot(K21,add=TRUE, legend=FALSE, lwd=2, col="blue")
legend("topleft",
       legend=c("K12","K21"),
       col=c("red","blue"), lwd=2, bty="n")
```



```r
plot(K13, legend=FALSE, ylab = "Kcross(r)", lwd=2, col="red", main="Cross-K btw 1-3 on network")
plot(K31, add=TRUE, legend=FALSE, lwd=2, col="blue")
legend("topleft",
       legend=c("K13","K31"),
       col=c("red","blue"), lwd=2, bty="n")
```

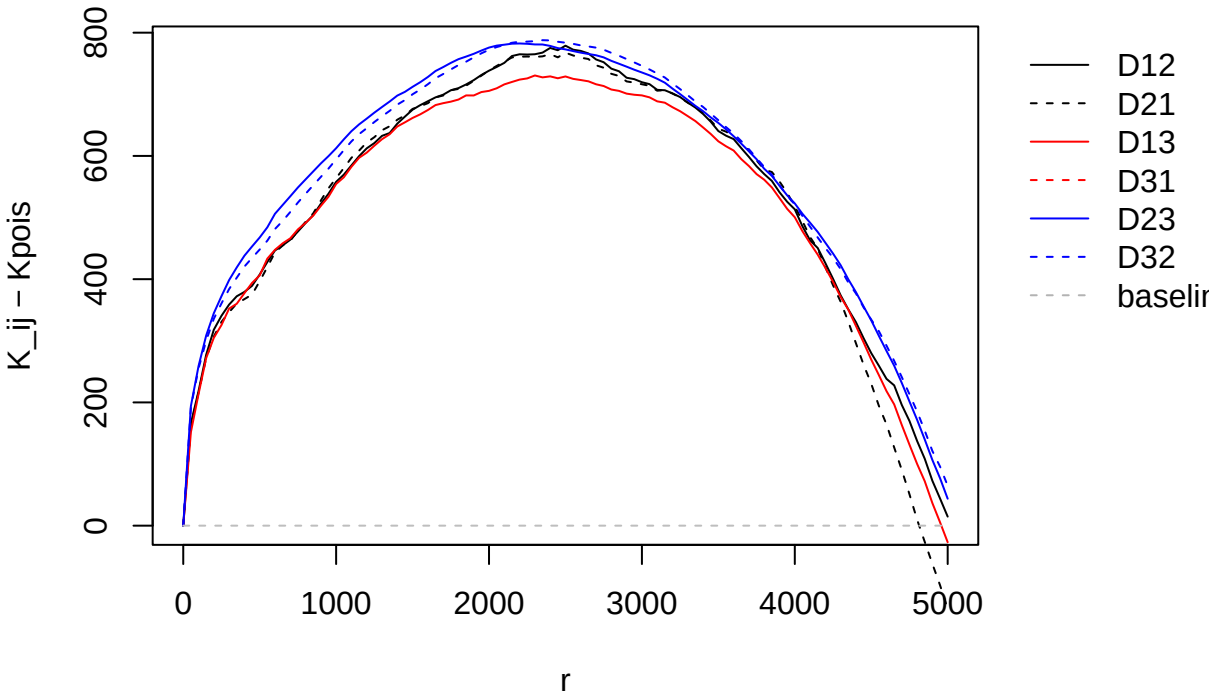



```r
plot(K23, legend=FALSE, ylab = "Kcross(r)", lwd=2, col="red", main="Cross-K btw 2-3 on network")

plot(K32, add=TRUE, legend=FALSE, lwd=2, col="blue")
legend("topleft",
       legend=c("K23","K32"),
       col=c("red","blue"), lwd=2, bty="n")
```




```
d12 <- K12$est - K12$r
d21 <- K21$est - K21$r
d13 <- K13$est - K13$r
d31 <- K31$est - K31$r
d23 <- K23$est - K23$r
d32 <- K32$est - K32$r

op <- par(no.readonly = TRUE)
par(mar = c(5, 5, 4, 6), xpd = NA)  # bigger margin


plot(K12$r, d12, type="l", ylab="K_ij - Kpois", xlab="r")
lines(K21$r, d21, lty=2, col = "black")

lines(K13$r, d13, lty=1, col = "red")   #
#lines(K31$r, d31, lty=2, col = "red")

lines(K23$r, d23, lty = 1, col = "blue")
lines(K32$r, d32, lty = 2, col = "blue")

lines(K12$r, K12$r-K23$r, lty = 2, col = "gray")

legend("topright", inset = c(-0.33, 0),
       legend = c("D12", "D21",

                  "D13", "D31",

                  "D23", "D32",
```

```
                 "baseline"),
       col    = c("black","black","red","red","blue","blue","grey70"),
       lty    = c(1,2,1,2,1,2,2),
       bty    = "n")
```




```
# K12in <- linearKcross.inhom(Y, i="sev1", j="sev2",lambdaI=NULL, lambdaJ=NULL, r=r_grid)
# K13in <- linearKcross.inhom(Y, i="sev1", j="sev3",lambdaI=NULL, lambdaJ=NULL, r=r_grid)
# K23in <- linearKcross.inhom(Y, i="sev2", j="sev3",lambdaI=NULL, lambdaJ=NULL, r=r_grid)
#
# plot(K12in, legend=FALSE, ylab = "Kincross(r)", lwd=2, main="Cross-K on network")
# plot(K13in, add=TRUE, legend=FALSE, lwd=2, col="blue")
# plot(K23in, add=TRUE, legend=FALSE, lwd=2, col="red")
# legend("topleft",
#        legend=c("K12in","K13in","K23in"),
#        col=c("black","blue","red"), lwd=2, bty="n")
```